

CSCI 1670/2670

Operating Systems

Staff

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The Course

- **Implement threads**
- **Learn about operating systems**
- **Implement one (Weenix)**
 - **file-system-related components (1670)**
 - **most of the rest of it (1690, 2670)**

Workload Components

- **Four moderate programming assignments**
 - **UThreads**: simple user-level threads package
 - **MThreads**: more sophisticated user-level threads package
 - **VFS**: system calls and high-level file system
 - **S5FS**: low-level file system
- **One large programming assignment: Weenix**
 - combines VFS and S5FS, as well as Processes, Drivers, and Virtual Memory
- **Four homeworks**
- **In-class TopHat questions**

Workload by Course

- **1670 Section 1**
 - quizzes: 10%
 - homeworks: 40%
 - moderate programming assignments: 50%
- **1670 Section 2**
 - homeworks: 44.44%
 - moderate programming assignments: 55.56%
- **1690**
 - large programming assignment: 100%

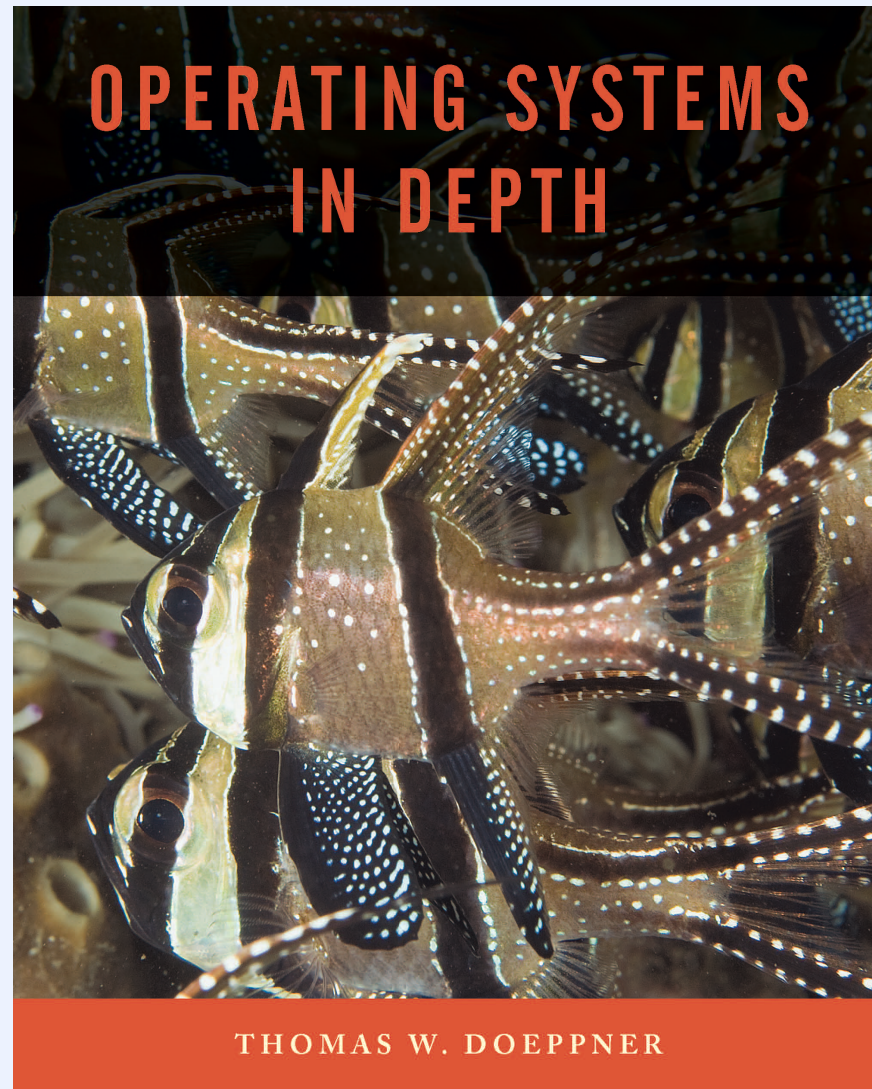
Workload by Course

- **2670 Section 1**
 - quizzes: 6.67%
 - homeworks: 26.68%
 - moderate programming assignments: 33.32%
 - large programming assignment: 33.33%
- **2670 Section 2**
 - homeworks: 28.58%
 - moderate programming assignments: 35.71%
 - large programming assignment: 35.71%

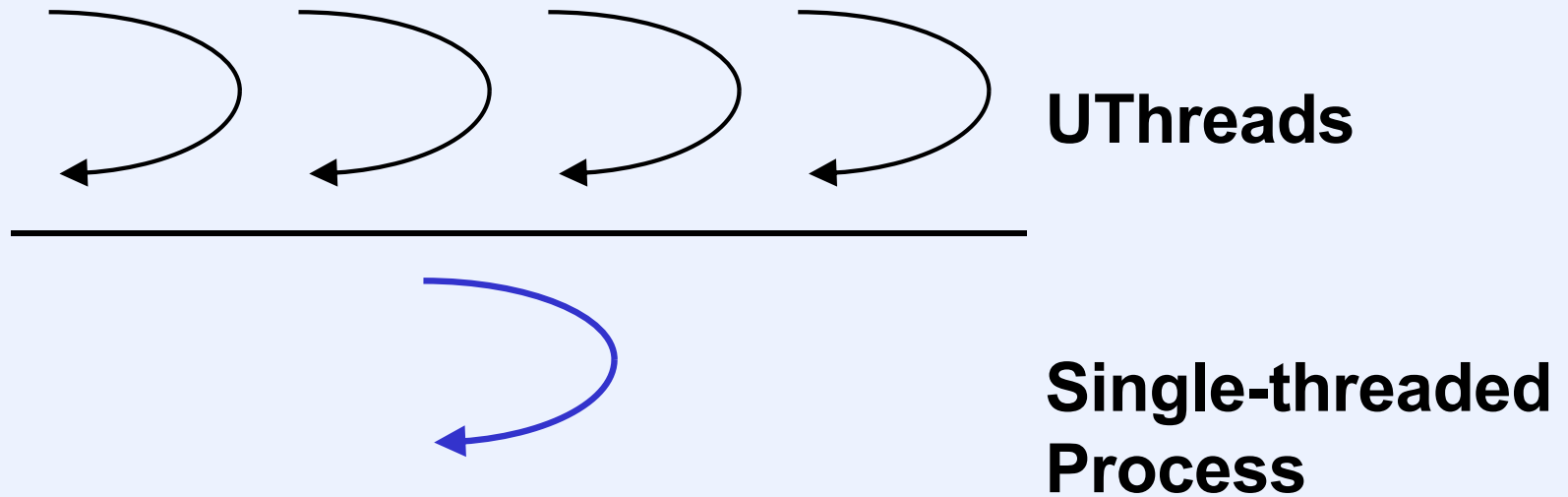
Skills Needed

- **Ability to write and debug largish programs in C with POSIX threads**
 - **CSCI 330/1330 or CSCI 300/1310 (you may want to review POSIX threads and signals material from 330 lectures)**
- **Basic computer architecture**
 - **CSCI 330/1330 or CSCI 300/1310**

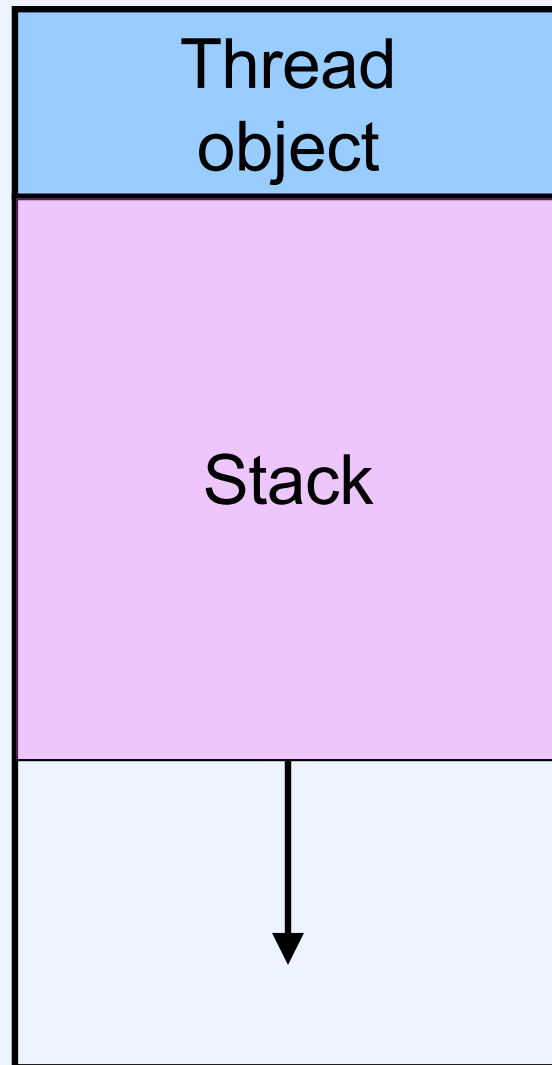
Textbook



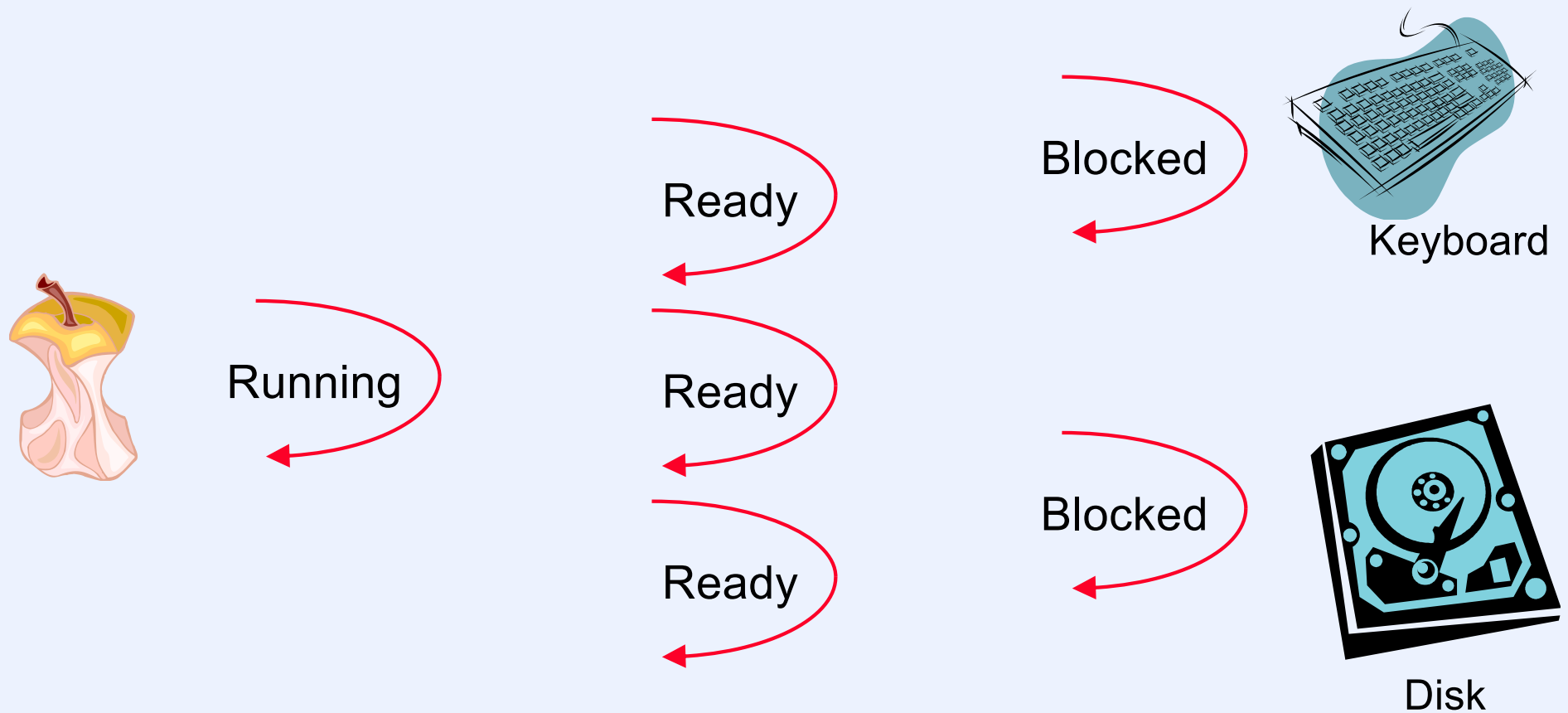
UThreads



UThread Implementation



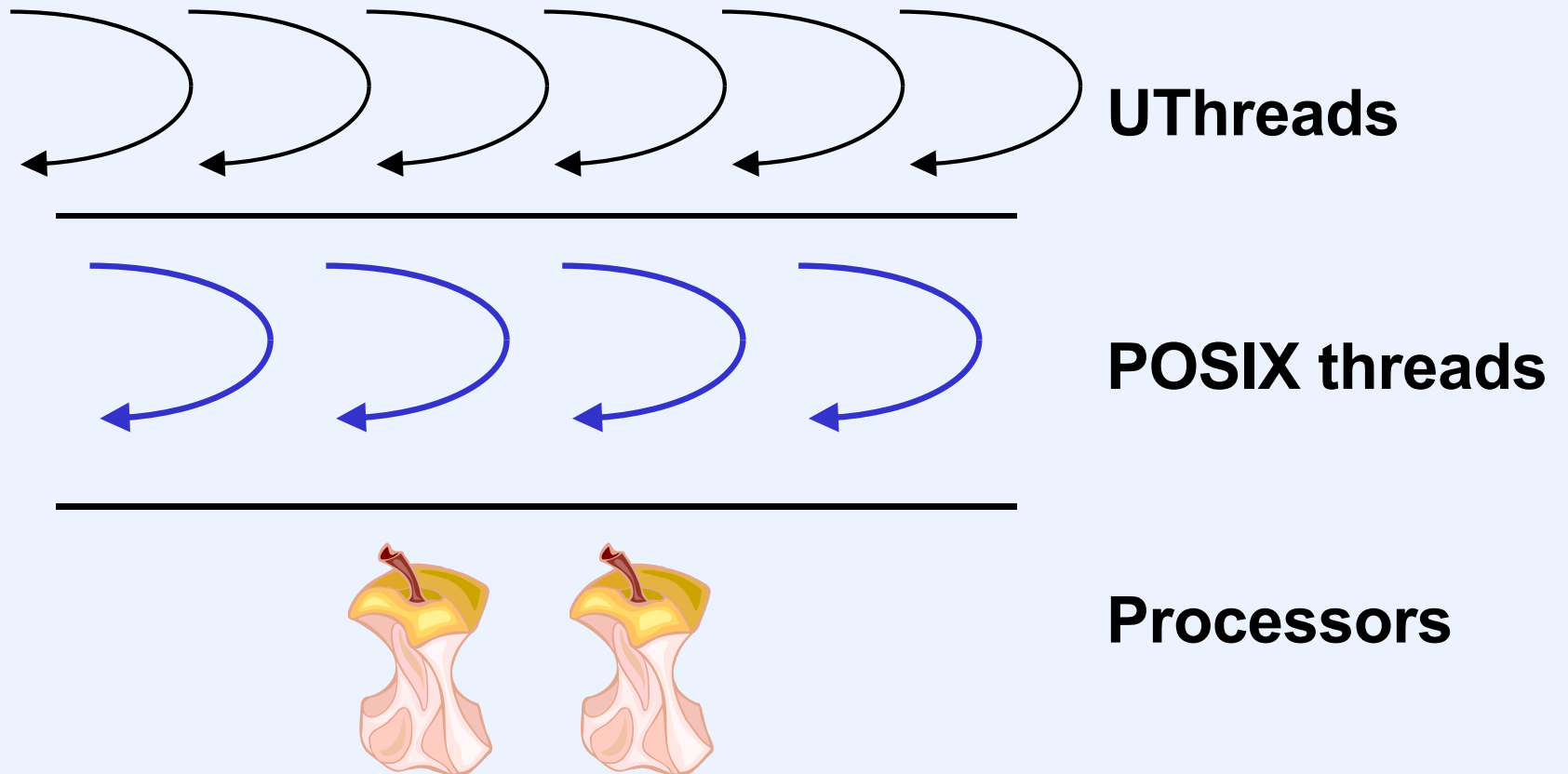
Multiplexing Threads (1)



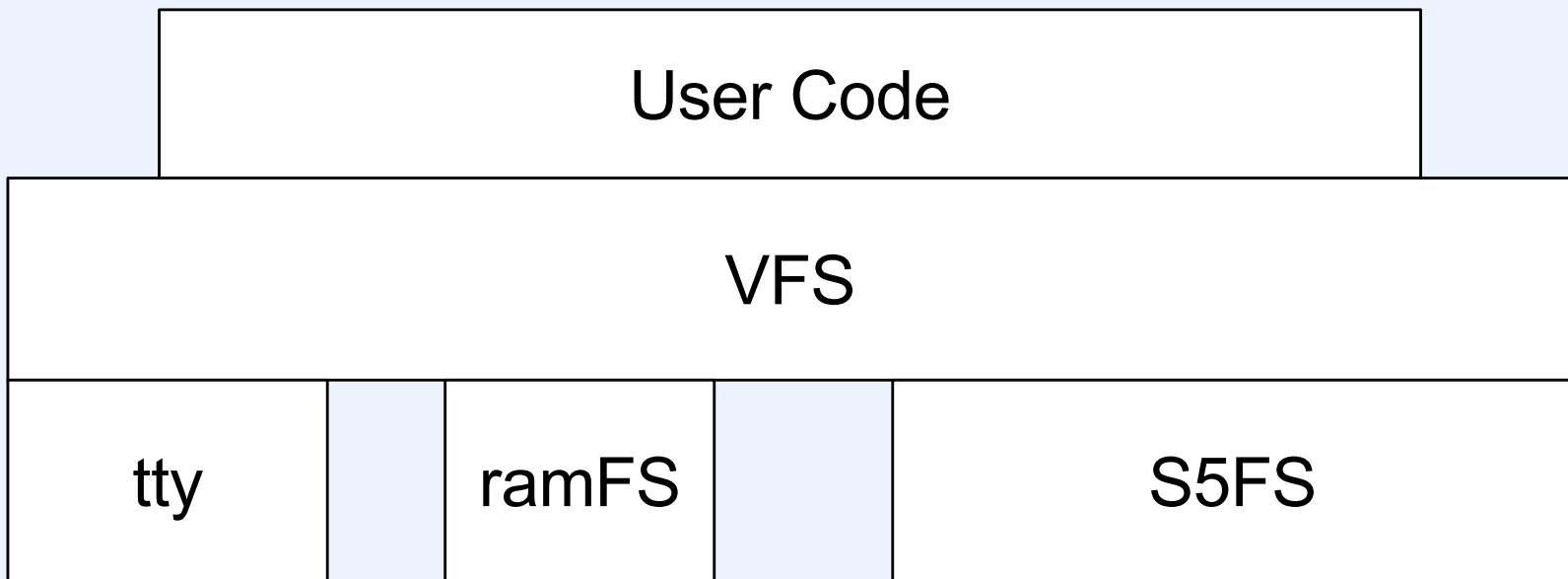
Multiplexing Threads (2)



MThreads

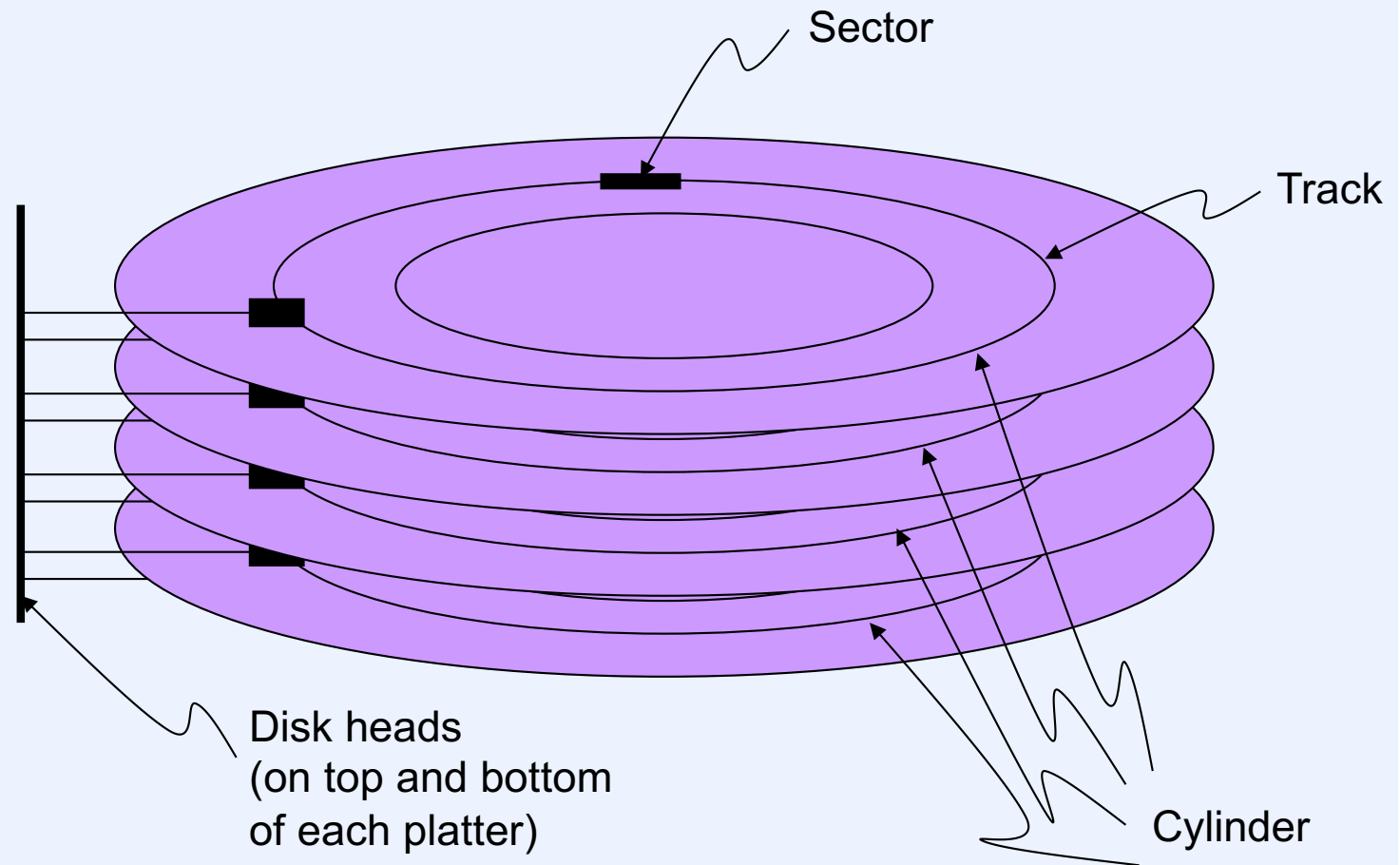


VFS

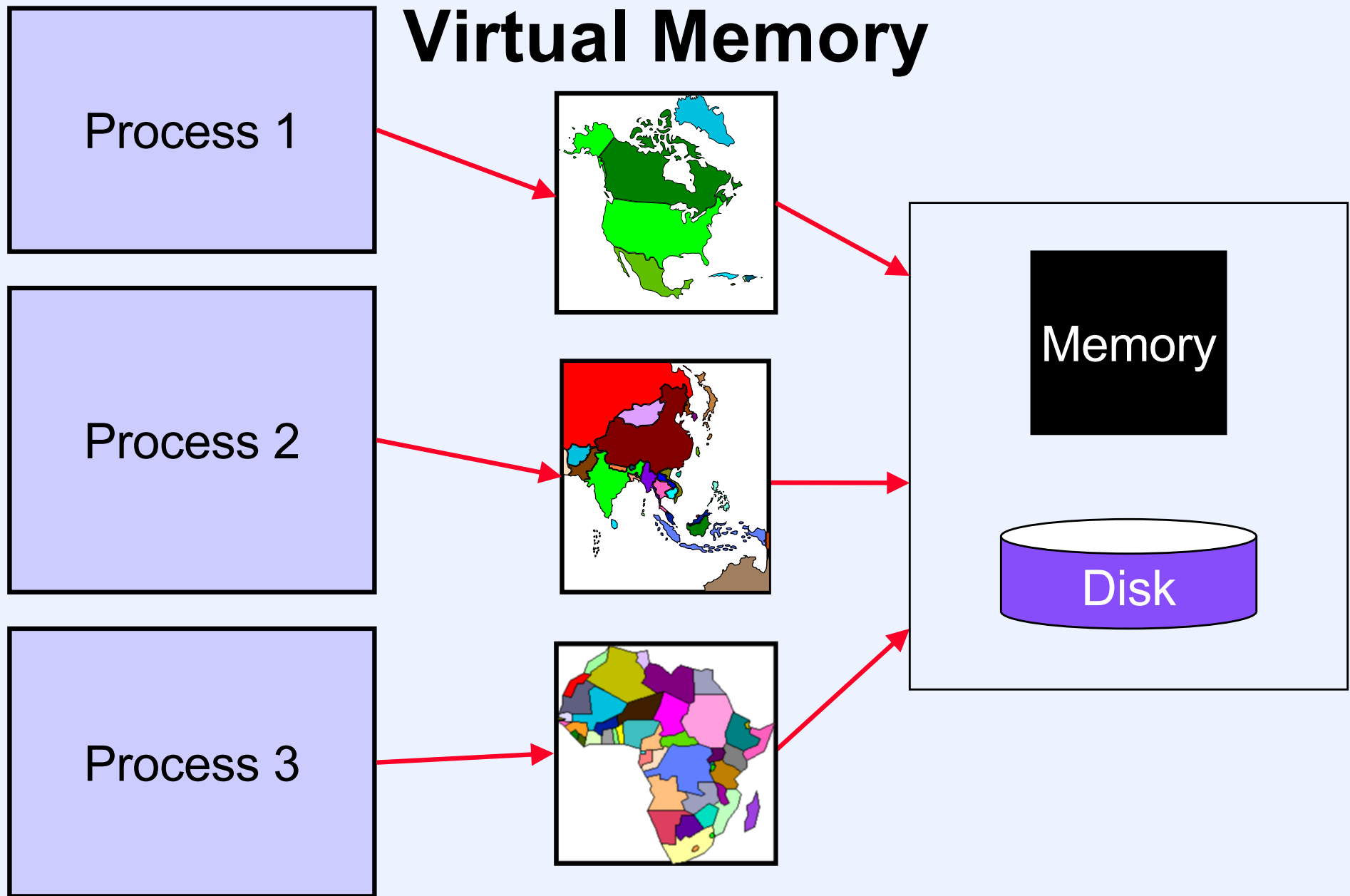


S5FS

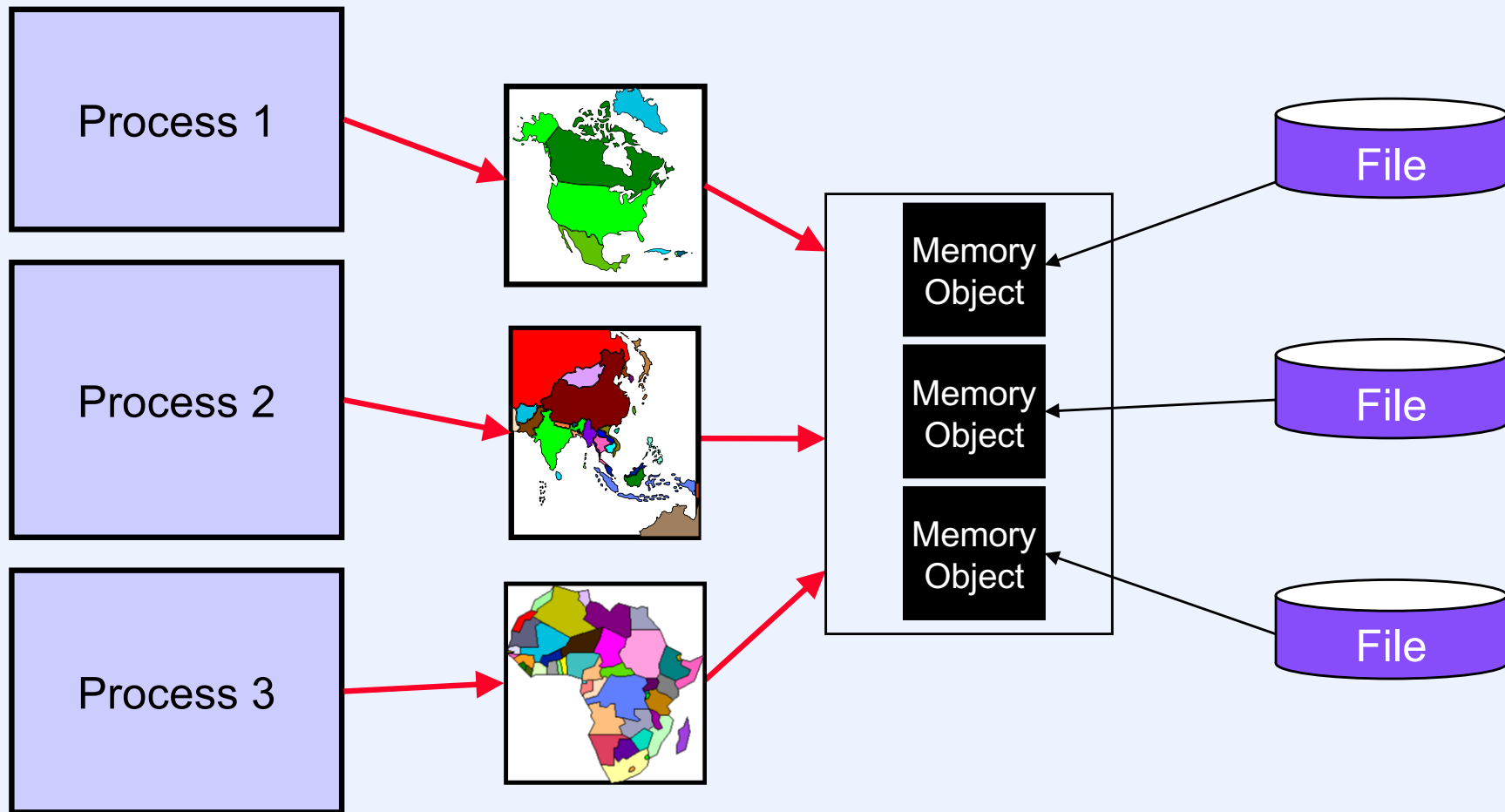
- It's on a disk!



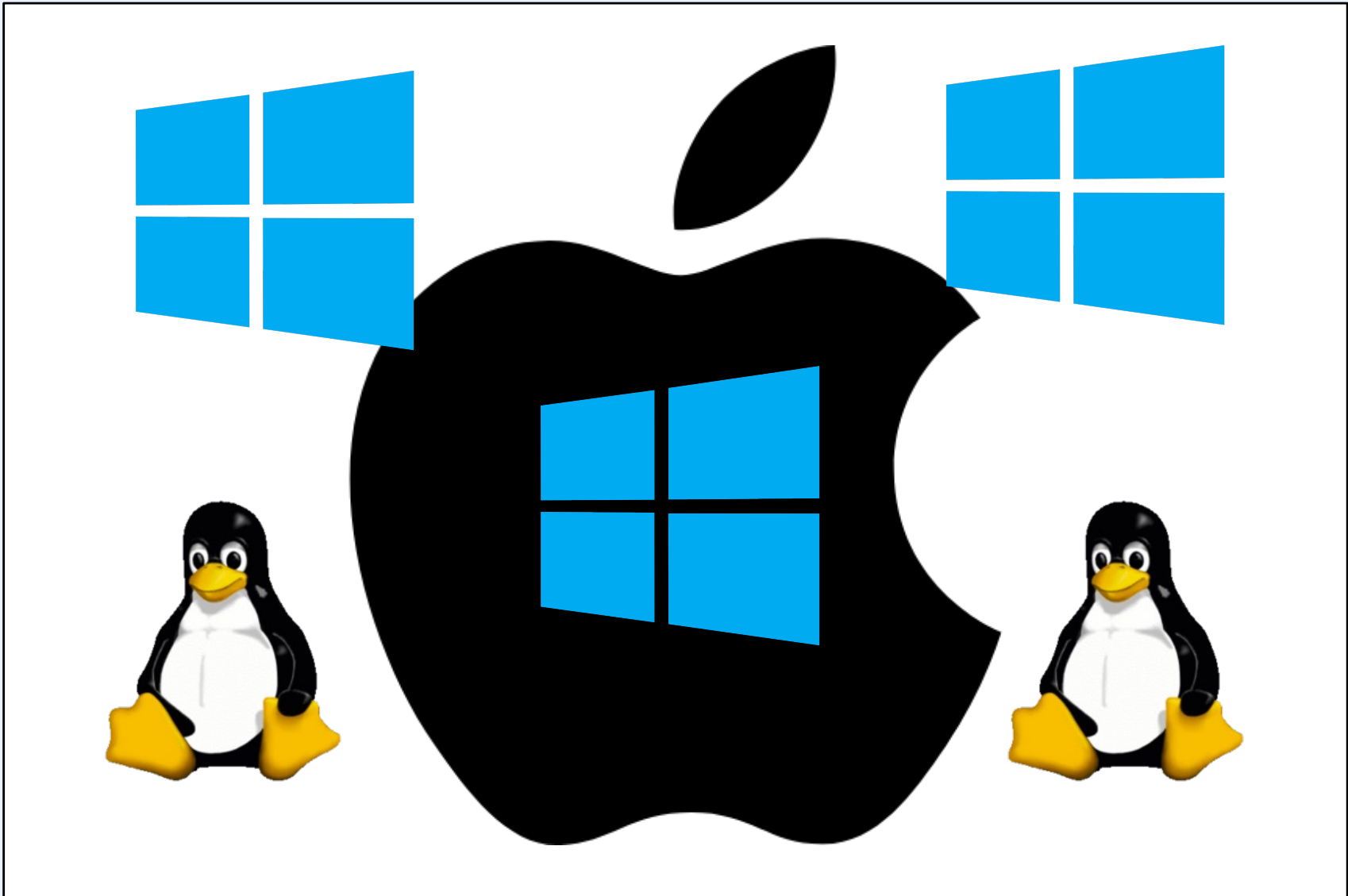
Virtual Memory



Virtual Memory + VFS



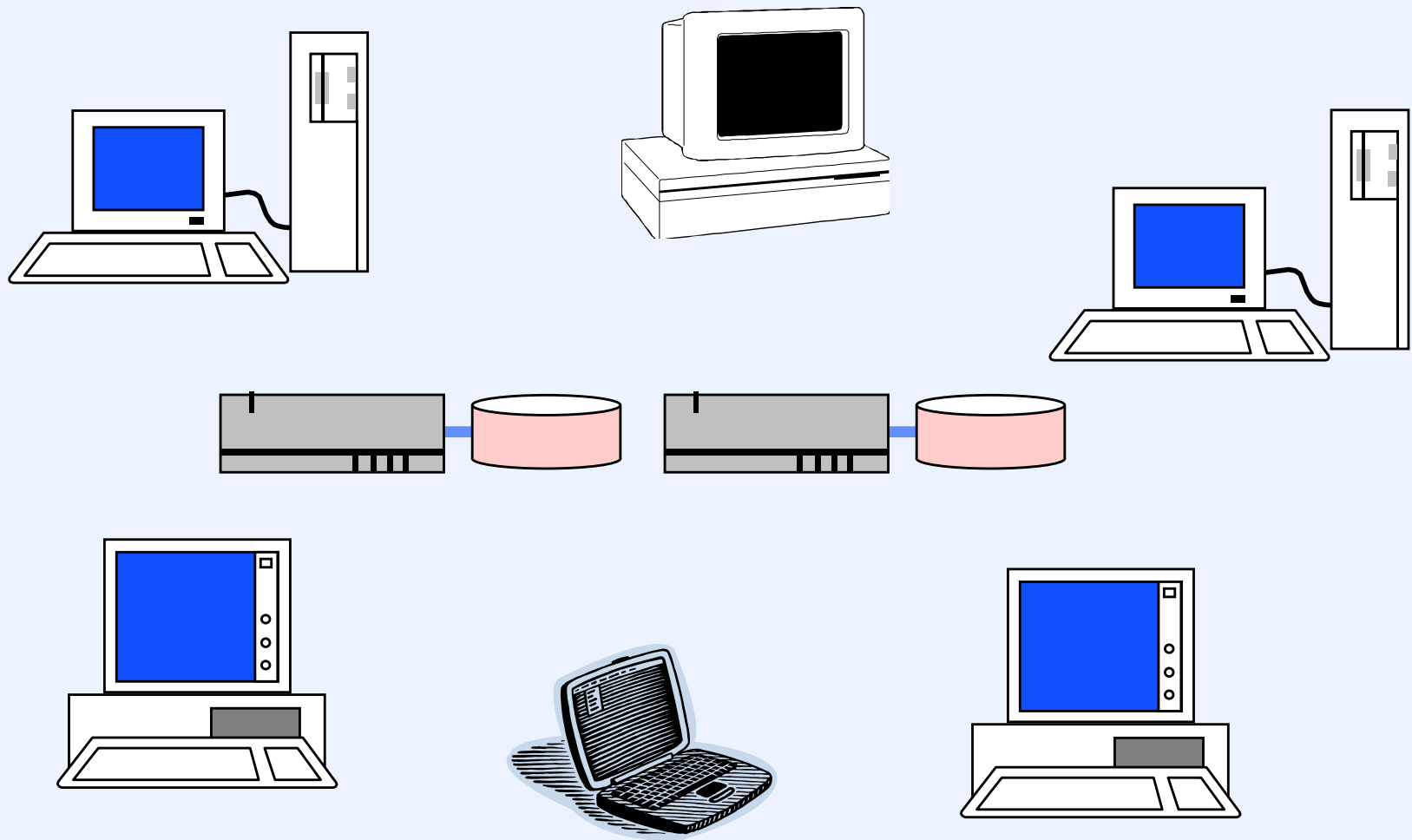
General Virtualization



OS-Level Security

- **Authentication**
 - who are you?
- **Authorization**
 - what are you allowed to do?
- **How is all this enforced?**
 - Windows/Mac/Linux
 - SELinux (security-enhanced Linux)
 - capability-based systems

Distributed File Systems



1960s OS Issues

- **Multiprogramming**
- **Time sharing**
- **Software complexity**
- **Security**

2025 OS Issues

- **Multiprogramming**
 - not just one computer, but server farms
- **Time sharing**
 - voice, video, sound, etc.
- **Software complexity**
 - a much bigger problem than could possibly be imagined in the '60s
- **Security**
 - ditto

In the Beginning ...

- **There was hardware**
 - processor
 - storage
 - card reader
 - tape drive
 - drum
- **And not much else**
 - no operating system
 - no libraries
 - no compilers

1950s

Commercial data processing

Scientific computing

IBM 701



OS:

Initially, none

IBM 650



OS:

none

1960s

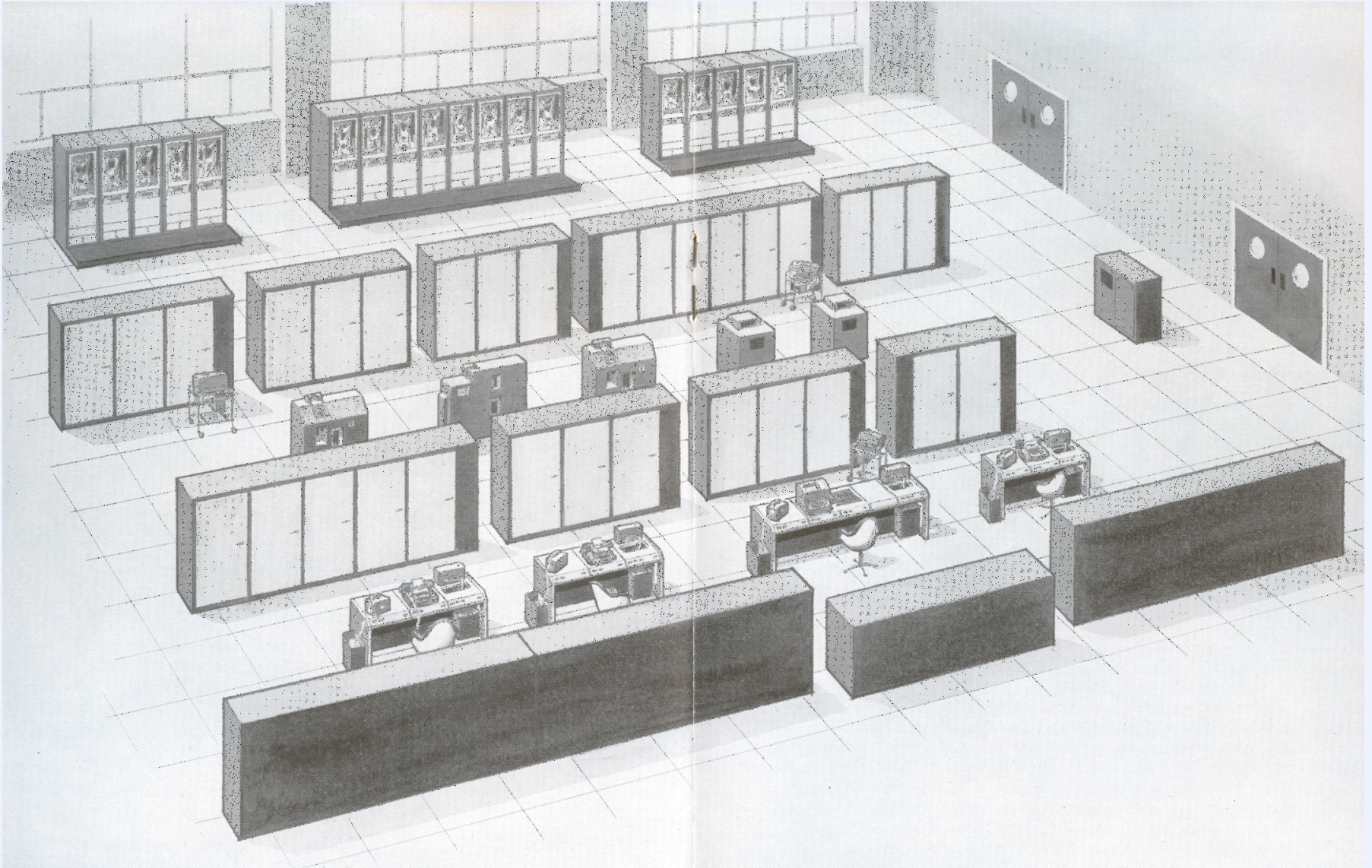
Commercial data processing

Scientific computing

Time sharing

Laboratory computing

Atlas Computer



IBM 7094



OS:

CTSS
(among others)

The IBM Mainframe



OS:

OS/360

Multics



DEC PDP-8



OS:

many:
ranging from
primitive to
interesting (a
multi-user time-
sharing system; a
virtual-machine
system)

Unix



History of Concurrency

- **Multiprogramming**
 - 1961, 1962: Atlas, B5000
 - 1965: OS/360 MFT, MVT
- **Timesharing**
 - 1961: CTSS (developed by MIT for IBM 7094); BBN time-sharing system for DEC PDP-1
 - mid 60s
 - Dartmouth Timesharing System (DTSS)
 - TOPS-10 (DEC)
 - late 60s
 - Multics (MIT, GE, Bell Labs)
 - Unix (Bell Labs)

History of Virtual Memory

- **1961: Atlas computer, University of Manchester, UK**
- **1962: Burroughs B5000**
- **1972: IBM OS/370**
- **1979: 3 BSD Unix, UC Berkeley**
- **1993: Microsoft Windows NT 3.1**
- **2000: Apple Macintosh OS X**

1970s

Commercial data processing

Scientific computing

Time sharing

Laboratory computing

Personal computing

Hobbyist computing

IBM's Dominance Continues



OS:

OS/370

Scientific Computing



OS:

**COS: single job
at a time**

Xerox Alto



OS:

**single-user,
single-
computation**

MIT'S ALTAIR 8800



OS:

none

CP/M

- **Control Program for Microcomputers**
 - 1974
 - first hobbyist OS
 - supported Intel 8080 and other systems
 - clear separation of architecture-dependent code
 - no multiprogramming
 - no protection

Apple II



OS:

**initially: none
later: similar
functionality as
CP/M (not much)**

Microsoft Enters the OS Business: Late 1970s

- **It's called ...**
 - **Xenix**
 - **a version of Unix**
 - **predominant version of Unix in the 1980s**
 - **used by MS internally into the 1990s**

VAX-11/780



OS:

**VMS
Unix**

Both:

- **time sharing**
- **virtual memory**
- **access protection**
- **concurrency**

1980s

Commercial data processing

Scientific computing

Time sharing

Laboratory computing

~~Personal~~ Professional computing

~~Hobbyist~~ Personal computing

Two OSes Take Off



Unix



MS-DOS

IBM PC



OS:

**PC-DOS
(aka MS-DOS)
(remarkably like
CP/M)**

The Computer Workstation



OS:

Aegis

supported:

- virtual memory
- distributed file system
- access protection
- concurrency

1990s

Commercial data processing

Scientific computing

High-end personal computing

Low-end personal computing

Toy Operating Systems

- **1987: Andrew Tanenbaum of Vrije Universiteit, Amsterdam, publishes *Operating Systems: Design and Implementation***
 - included is source code for a complete, though toy, operating system: Minix, sort of based on Unix
- **1991: Linus Torvalds buys an Intel 386 PC**
 - MS-DOS doesn't support all its features (e.g., memory protection, multi-tasking)
 - rewrites Minix to support all this
- **January 1992: Torvalds releases Linux 0.12**
- **January 1992: Tanenbaum declares Linux obsolete**

Late '80s/Early '90s

- **1988: Most major Unix vendors get together and form OSF to produce a common Unix: OSF/1, based on IBM's AIX**
 - **1989: Microsoft begins work on NT**
 - **1990: OSF abandons AIX, restarts with Mach**
 - **1991: OSF releases OSF/1**
 - **1992: Sun releases Solaris 2**
 - many SunOS (Solaris 1) programs are broken
 - **1993: All major players but DEC have abandoned OSF/1**
 - **1993: Microsoft releases Windows NT 3.1**
 - **1994: Linux 1.0 released**
-

Late '90s

- **1996: DEC renames its OSF/1 “Digital Unix”**
- **1996: Microsoft releases Windows NT 4**
- **1996: Linux 2.0 released**
- **1998: DEC is purchased by Compaq; “Digital Unix” is renamed “Tru64 Unix”**
- **1999: Sun’s follow-on to Solaris 2.6 is called Solaris 7**

2000s

Commercial data processing

Scientific computing

Personal computing

Gadgets/Smartphones

The '00s Part 1

- **2000: Microsoft releases Windows 2000 and Windows Me**
- **2000: Linux 2.2 is released**
- **2000: IBM “commits” to Linux (on servers)**
- **2001: Apple releases Mac OS X, based on Unix and Mach (sort of like OSF/1)**
- **2001: Linux 2.4 is released**
- **2001: Microsoft releases Windows XP**
- **2002: Compaq is purchased by HP**
- **2003: SCO claims their code is in Linux, sues IBM; IBM countersues**
 - **August 10, 2007: judge rules that SCO is not the rightful owner of the Unix copyright, Novell is**
 - **Novell says there is no Unix in Linux**
 - **September 2007: SCO files for Chapter 11 bankruptcy protection**

The '00s Part 2

- **2004: Linux 2.6 is released**
 - **2005: IBM sells PC business to Lenovo**
 - **July 2005: Microsoft announces Windows Vista**
 - **January 2007: Microsoft releases Windows Vista**
 - **Later in 2007: Microsoft starts hinting at Windows 7**
 - **June 2007: Apple releases iOS for iPhone**
 - **April 2009: Oracle announces purchase of Sun Microsystems**
 - **July 2009: Google announces Chrome OS**
 - **October 2009: Microsoft releases Windows 7**
-

The '10s Part 1

- **January 2011: Microsoft announces Windows 8**
- **June 2011: first products shipped running Chrome OS**
- **April 2011: Linux 3.0 released**
- **July 2011: Apple drops “Mac” prefix for its OS: it’s now “OS X”**
 - one million copies of OS X Lion sold on first day of release
- **October 2011: deaths of both Dennis Ritchie and Steve Jobs**
 - “Dennis Ritchie: The Shoulders Steve Jobs Stood On”
 - Wired Magazine, 10/13/2011
- **October 2012: Microsoft releases Windows 8**

The '10s Part 2

- **September 2014: Microsoft Announces Windows 10**
 - Windows 9 skipped because it might be confused with Windows 95
- **October 2014: Apple releases OS X 10.10 Yosemite**
 - doesn't work with Brown University wi-fi
 - November 2014 release of 10.10.1 doesn't fix it ...
- **April 2015: Linux 4.0 released**
- **July 2015: Microsoft Releases Windows 10**
 - Brown's CIS says "don't switch yet"
 - August 2015: Brown's card-access system switched to Windows 10
 - August 2015: Brown's card access fails university-wide
- **September 2015: Apple releases OS X 10.11 El Capitan**
 - Brown's CIS says "don't switch yet"

The '10s Part 3

- **September 2016: Apple releases OS X 10.12 Sierra**
 - Brown CIS says don't switch
 - Brown CS tstaff says printing is broken
- **December 2016: Apple releases OS X 10.12.2**
 - twd switches
 - his D-Link router no longer supported printing
 - his Logitech mouse's scroll wheel scrolled window in random directions
- **January 2018: Meltdown and Spectre vulnerabilities disclosed**
 - end of computing (if not civilization) as we know it?

The '10s Part 4

- **September 2018: Apple releases OS X 10.14 Mojave**
 - no one switches
- **October 2018: Apple releases OS X 10.14.1**
 - two switches
 - PowerPoint presentations in CS33 randomly switch to random photos
- **March 2019: Linux 5.0 released**
- **October 2019: Apple releases OS X 10.15 Catalina**

The '20s Part 1

- **November 2020: Apple releases OS X 11.0 Big Sur**
 - twd upgraded his laptop
 - CIS-supported antivirus software won't let machine shut down
 - fixed in 11.1
- **October 2021: Apple releases OS X 12.0 Monterey**
 - January 2022: OIT (formerly CIS) finally says it's ok to use it
 - twd bought a new laptop that came with Monterey (12.0.1) installed
 - twd's 7-year-old iMac was too old to run it
- **October 2022: Apple releases OS X 13.0 Ventura**
 - CSCI 330/1330 students who installed it regretted it

The '20s Part 2

- **September 2023: Apple announces OS X 14.0 Sonoma**
 - OIT says many things don't work
 - twd waits till 14.2.1 before switching
 - Logitech mouse's tracking speed can't be adjusted
- **September 2024: Apple announces macOS 15.0 Sequoia**
 - macOS reintroduced to differentiate it from iOS
 - twd gets new desktop with 15.0
 - no major problems!
 - now running 15.1.1

Friday

- **Implementing threads**
 - now is a good time to review POSIX threads